

Analysis of Variance as Linear Model Theory

Design matrices, projections, estimability, sums of squares, and F tests

$$Y = X\beta + \varepsilon$$



mean

factor

error

Learning objectives

ANOVA is not a separate method: it is a structured normal linear model.

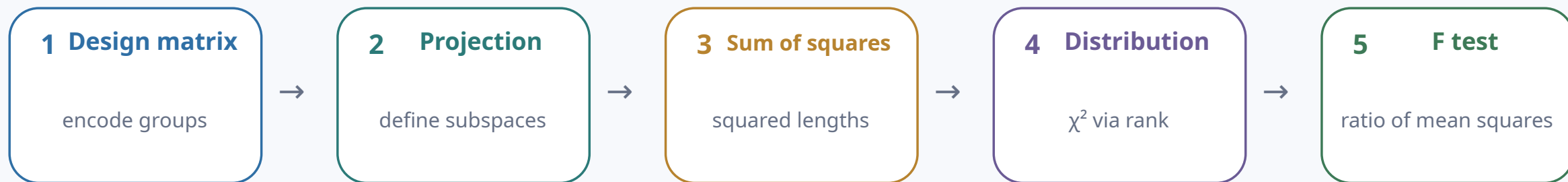
ROADMAP

- Write one-way and two-way fixed-effects ANOVA models as $Y = X\beta + \varepsilon$.
- Explain why common effects parameterizations are rank deficient and identify estimable functions.
- Derive ANOVA sums of squares as quadratic forms in projection matrices.
- Obtain degrees of freedom from ranks and exact F tests from Cochran's theorem.
- Formulate factor and interaction hypotheses as general linear hypotheses.
- Explain why balanced designs are especially simple and what changes in unbalanced designs.

The big picture

ANOVA reorganizes linear-model geometry around factors and contrasts.

BIG PICTURE



Core idea: an ANOVA “source of variation” is a model subspace—or an increment between nested model subspaces.

01

One-way ANOVA

From group means to projection matrices and exact F inference.

One-way ANOVA setup

NOTATION

Suppose factor A has g levels. Observation j in group i follows

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2),$$
$$N = \sum_{i=1}^g n_i$$

with $i = 1, \dots, g$ and $j = 1, \dots, n_i$. Total sample size:

Scientific question

Are all group means equal, or does the factor explain systematic variation in the response?

Linear-model question

Does enlarging the mean subspace from one common mean to g group-specific means significantly reduce residual variation?

Cell-means parameterization

FULL-RANK MODEL

Stack the N observations into Y and define $\beta = (\mu_1, \dots, \mu_g)^T$

The design matrix has one indicator column per group:

row	G1	G2	G3	...
group 1	1	0	0	...
group 1	1	0	0	...
group 2	0	1	0	...
group 3	0	0	1	...

Rank

If every group has at least one observation, rank(X)=g. The group means μ_i are individually identifiable.

This parameterization is often the cleanest for theory.

Effects parameterization and rank deficiency

IDENTIFIABILITY

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$$

Now $\beta = (\mu, \alpha_1, \dots, \alpha_g)^T$ has $g+1$ entries, but the mean space still has dimension only g .

Nonuniqueness

$$(\mu, \alpha_i) \mapsto (\mu + c, \alpha_i - c)$$

What remains unique?

$$\mu_i = \mu + \alpha_i, \quad \alpha_i - \alpha_j$$

Constraints such as $\sum \alpha_i = 0$ choose a parameterization; they do not create information.

Two parameterizations, one model space

SAME $\mathcal{C}(X)$

Cell means

$$\beta = (\mu_1, \dots, \mu_g)$$

X has g columns and rank g .

Each μ_i is estimable.

Overall mean + effects

$$\beta = (\mu, \alpha_1, \dots, \alpha_g)$$

X has $g+1$ columns but rank g .

Individual μ and α_i depend on the chosen constraint.

$$\mathcal{C}(X_{\text{cell means}}) = \mathcal{C}(X_{\text{effects}})$$

Fitted values, residuals, SSE, and tests of estimable hypotheses are identical.

Estimability in the effects model

ESTIMABILITY

A linear function $c^T\beta$ is estimable exactly when it is invariant to the null-space direction.

$$h = (1, -1, -1, \dots, -1)^T, \qquad Xh = 0$$

Thus estimability requires $c^Th = 0$.

Quantity	Estimable?	Reason
μ	No	changes along h
α_i	No	changes along h
$\mu + \alpha_i$	Yes	group mean
$\alpha_i - \alpha_j$	Yes	contrast

The ANOVA null hypothesis is a general linear hypothesis

HYPOTHESIS

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_g$$

Equivalent contrast form: $H_0 : C\mu = 0, \quad \text{rank}(C) = g - 1$

One convenient C compares each later group with group 1:

Degrees of freedom

The hypothesis contains $g-1$ linearly independent restrictions.

ANOVA as a comparison of nested mean spaces

NESTED MODELS

Reduced model

One common mean: $Y = 1 \mu + \varepsilon$
 $\text{rank} = 1$

Full model

Separate group means: $Y = X_F \mu + \varepsilon$
 $\text{rank} = g$

Added directions

$C(X_R) \subset C(X_F)$
 $\dim = g - 1$

$$SS_A = SSE_R - SSE_F = Y^T (H_F - H_R) Y$$

“Between-groups variation” is the extra sum of squares gained by allowing different means.

The classical decomposition is a projection identity

SUM OF SQUARES

Let $J = \mathbf{1}\mathbf{1}^T/N$ project onto the grand mean and H project onto the group-mean space.

$$I - J = (H - J) + (I - H)$$

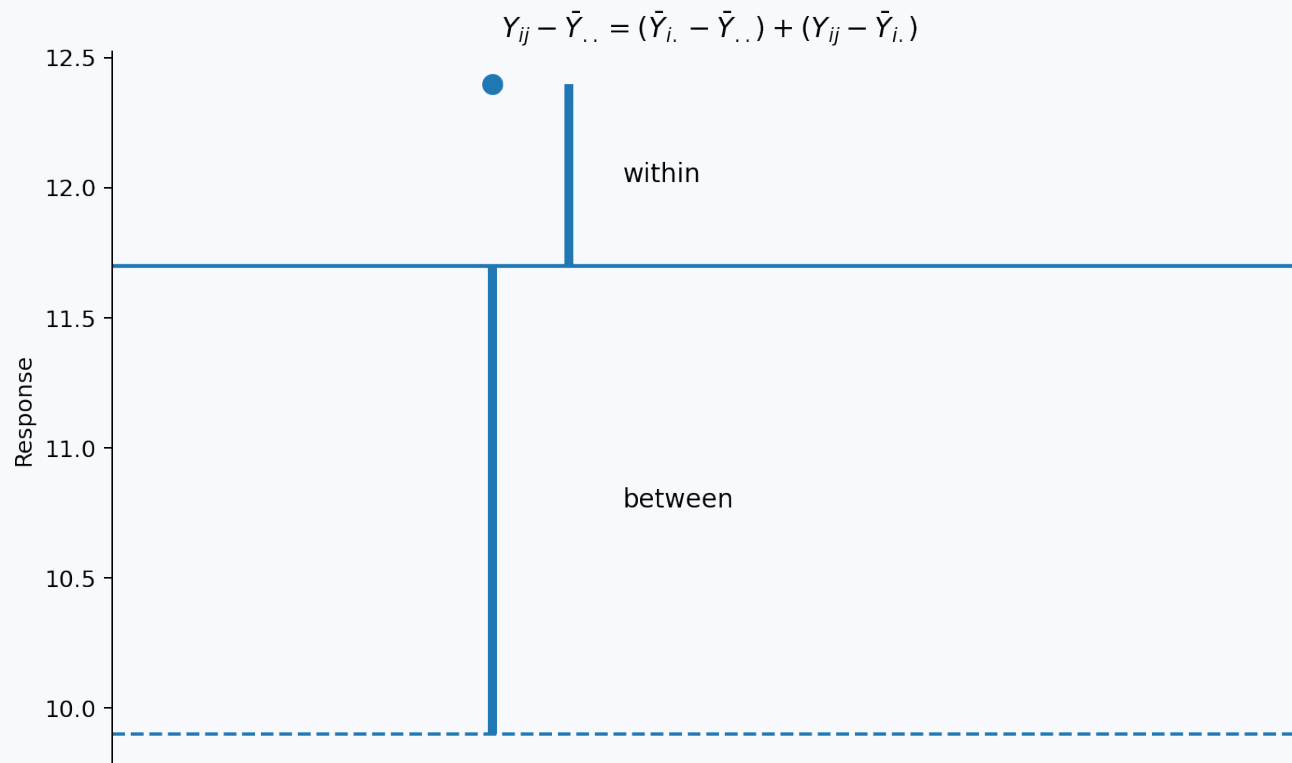
The two pieces are orthogonal projections:

$$(H - J)(I - H) = 0$$

Total = between groups + within groups

One-way ANOVA in the data space

GEOMETRY



Between groups

Distance from the grand-mean fit to the group-mean fit.

Within groups

Distance from the observations to their fitted group means.

Orthogonality makes the squared lengths add.

Classical one-way sums of squares

FORMULAS

$$SS_A = \sum_{i=1}^g n_i (\bar{Y}_{i.} - \bar{Y}_{..})^2 \quad SS_E = \sum_{i=1}^g \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{i.})^2$$

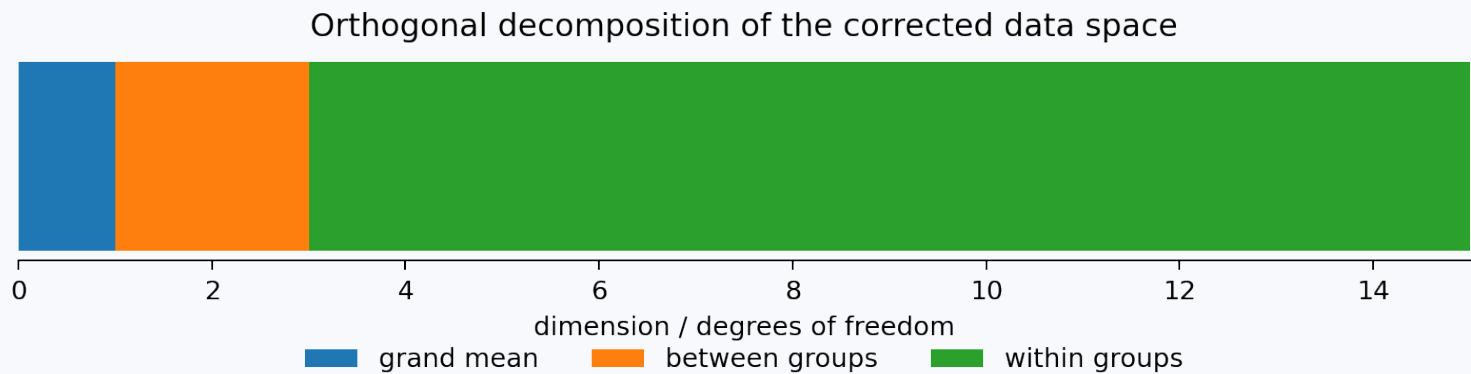
$$SS_T = \sum_{i=1}^g \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{..})^2$$

$$SS_T = SS_A + SS_E$$

These scalar formulas are coordinate representations of the projection identity.

Degrees of freedom are ranks

RANK



Component	Projection	rank
Grand mean	J	1
Factor A	H-J	$g-1$
Error	I-H	$N-g$
Corrected total	I-J	$N-1$

$$N - 1 = (g - 1) + (N - g)$$

The familiar ANOVA df decomposition is literally a dimension decomposition.

Cochran's theorem gives the null distributions

DISTRIBUTION

Under H_0 , the true mean lies in $C(J)$, so $(H-J)\mu = 0$.

$$\frac{SS_A}{\sigma^2} \sim \chi_{g-1}^2,$$

$$\frac{SS_E}{\sigma^2} \sim \chi_{N-g}^2$$

$$SS_A \perp SS_E$$

Why χ^2 ?

$H-J$ and $I-H$ are symmetric idempotent matrices. Their ranks become degrees of freedom.

Why independent?

The two projections are orthogonal. Cochran's theorem turns orthogonal Gaussian components into independent sums of squares.

The one-way F test

F TEST

$$MS_A = \frac{SS_A}{g - 1}, \quad MSE = \frac{SS_E}{N - g}$$

$$F = \frac{MS_A}{MSE} = \frac{SS_A / (g - 1)}{SS_E / (N - g)} \sim F_{g-1, N-g} \quad \text{under } H_0$$

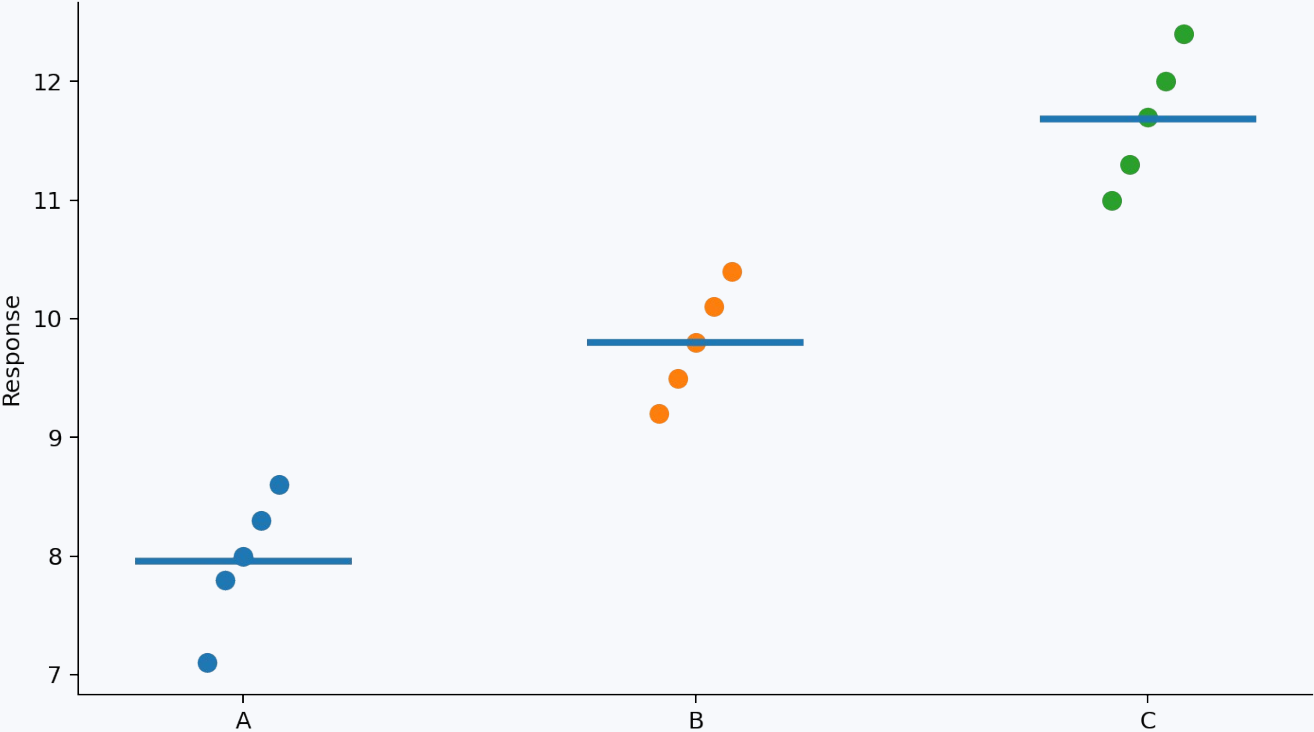
Interpretation Compare average squared signal in the factor subspace with average squared noise in the residual subspace.

Large F means the added group-mean directions explain much more variation than residual noise alone would predict.

Worked example: three treatment groups

EXAMPLE

One-way ANOVA: observations and group means



Group	Observations	Mean
A	7.1, 7.8, 8.0, 8.3, 8.6	7.96
B	9.2, 9.5, 9.8, 10.1, 10.4	9.80
C	11.0, 11.3, 11.7, 12.0, 12.4	11.68

Grand mean = 9.813

N = 15, g = 3

Worked example: ANOVA calculations

EXAMPLE

$$SS_A = 34.597, \quad SS_E = 3.420, \quad SS_T = 38.017$$

Source	df	SS	MS	F
Treatment	2	34.597	17.299	60.70
Error	12	3.420	0.285	
Total	14	38.017		

F = 60.70 with df (2,12); p ≈ 5.3×10⁻⁷

The three fitted means occupy a much richer mean subspace than the common-mean model.

Contrasts ask focused questions among group means

CONTRASTS

A contrast is $\mathbf{L} = \sum_{i=1}^g c_i \mu_i$, $\sum_{i=1}^g c_i = 0$

Examples

$$\mu_1 - \mu_2$$

$$(\mu_1 + \mu_2)/2 - \mu_3$$

linear trend contrasts for ordered levels

Estimator and variance

$$\hat{L} = \sum c_i \bar{Y}_i$$

$$\text{Var}(\hat{L}) = \sigma^2 \sum c_i^2 / n_i$$

The omnibus F test asks whether all $g-1$ independent contrast directions are zero simultaneously.

Inference for a single contrast

T TEST

$$t = \frac{\hat{L} - L_0}{\sqrt{MSE \sum_i c_i^2 / n_i}} \sim t_{N-g} \quad \text{under } H_0 : L = L_0$$

Equivalently, the 1-df sum of squares for testing $L = 0$ is

$$SS_L = \frac{\hat{L}^2}{\sum_i c_i^2 / n_i}$$

$$F_L = \frac{SS_L}{MSE} = t^2$$

A one-dimensional contrast test is just a 1-df general linear F test.

Orthogonal contrasts can partition the factor sum of squares

BALANCED DESIGNS

For equal group sizes n , two contrasts c and d are orthogonal when

$$\sum_{i=1}^g c_i d_i = 0$$

A complete set of $g-1$ mutually orthogonal contrasts spans the factor subspace.

$$SS_A = SS_{L_1} + \cdots + SS_{L_{g-1}}$$

This partition is especially clean because the corresponding projection directions are perpendicular.

One-way ANOVA is regression with indicator variables

EQUIVALENCE

Choose group 1 as a reference and fit

$$Y_i = \beta_0 + \beta_2 D_{2i} + \cdots + \beta_g D_{gi} + \varepsilon_i$$

Parameter	Meaning
β_0	mean of group 1
β_2	$\mu_2 - \mu_1$
β_3	$\mu_3 - \mu_1$
...	...

Same test
 $H_0: \beta_2 = \cdots = \beta_g = 0$ is exactly the one-way ANOVA test of equal means.

ANOVA and regression differ in interpretation and coding—not in underlying theory.

02

Two-way fixed-effects ANOVA

Main effects, interactions, balanced orthogonality, and general linear hypotheses.

Two factors: notation and cell means

SETUP

Factor A has a levels; factor B has b levels; r_{ij} observations may occur in cell (i,j) .

$$Y_{ijk} = \mu_{ij} + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \stackrel{iid}{\sim} N(0, \sigma^2)$$

Cell-means model

One parameter μ_{ij} per observed cell. With all ab cells present, the mean space has dimension ab .

Factor-effects model

Rewrite cell means using an overall mean, A effects, B effects, and interaction effects.

The two-way effects model

PARAMETERIZATION

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

Interpretation:

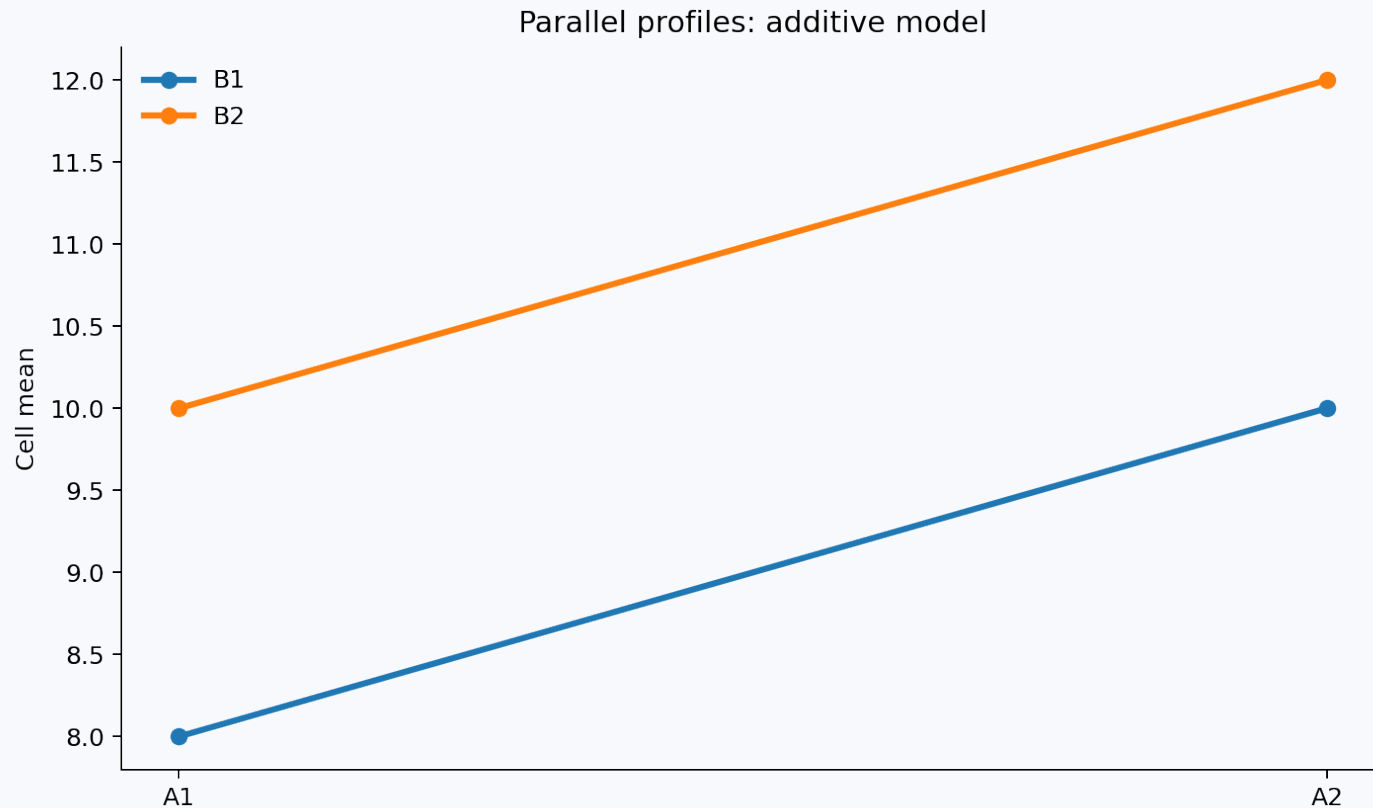
- μ : overall location.
- α_i : systematic change associated with level i of factor A.
- β_j : systematic change associated with level j of factor B.
- $(\alpha\beta)_{ij}$: departure from additivity for cell (i,j) .

Important

Without constraints, this parameterization is highly redundant. The cell means μ_{ij} are the identifiable objects.

No interaction means additivity

INTERACTION



Additive model

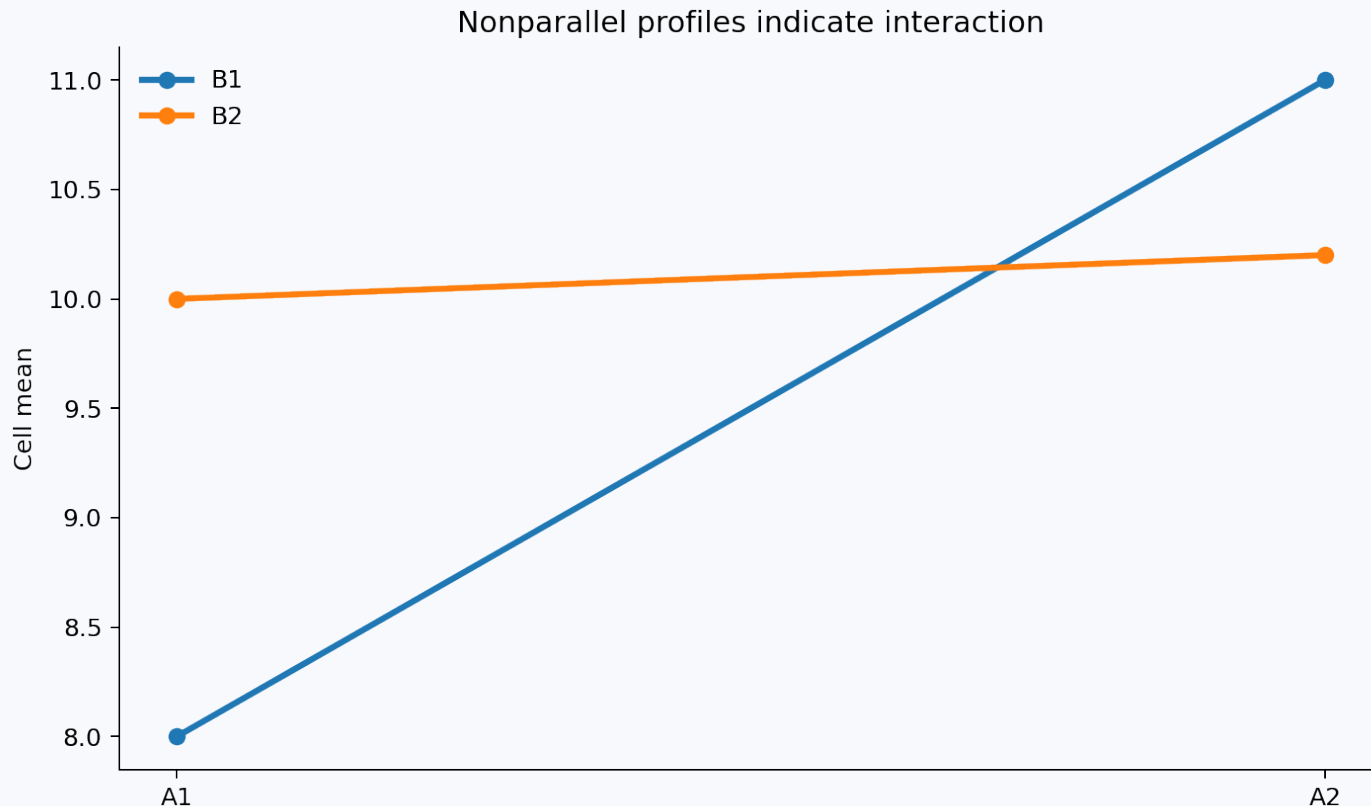
The difference between B levels is the same at every A level. Profile lines are parallel.

$$\mu_{ij} = \mu + \alpha_i + \beta_j$$

No extra cell-specific departure remains.

Interaction means the effect of one factor depends on the other

INTERACTION



Interaction

Nonparallel profiles indicate that simple effects differ across levels of the other factor.

$$(\alpha\beta)_{ij} \neq 0 \text{ for some cells}$$

With strong interaction, marginal main effects can be scientifically misleading.

Rank and constraints in the two-way effects model

IDENTIFIABILITY

The raw effects parameterization contains $1 + a + b + ab$

symbols, but with all cells present the saturated mean space has dimension only ab .

A common identifying convention is

$$\sum_i \alpha_i = 0, \quad \sum_j \beta_j = 0, \quad \sum_i (\alpha\beta)_{ij} = 0, \quad \sum_j (\alpha\beta)_{ij} = 0$$

These constraints make parameters unique; estimable hypotheses must not depend on which valid constraint system is chosen.

Balanced two-way ANOVA gives orthogonal model pieces

BALANCED DESIGN

Assume r observations in every cell.

$$I - J = P_A + P_B + P_{AB} + P_E$$

For a balanced design these projections are mutually orthogonal.

A

df = $a - 1$

B

df = $b - 1$

A×B

df = $(a - 1)(b - 1)$

Error

df = $ab(r - 1)$

$$abr - 1 = (a - 1) + (b - 1) + (a - 1)(b - 1) + ab(r - 1)$$

Balanced two-way sums of squares

FORMULAS

$$SS_A = br \sum_i (\bar{Y}_{i..} - \bar{Y}_{...})^2 \quad SS_B = ar \sum_j (\bar{Y}_{.j.} - \bar{Y}_{...})^2$$

$$SS_{AB} = r \sum_i \sum_j (\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...})^2$$

$$SS_E = \sum_{i,j,k} (Y_{ijk} - \bar{Y}_{ij.})^2$$

$$SS_T = SS_A + SS_B + SS_{AB} + SS_E$$

Balanced two-way ANOVA table

ANOVA TABLE

Source	df	Sum of squares	Mean square	F
A	$a-1$	SS_A	MS_A	MS_A/MSE
B	$b-1$	SS_B	MS_B	MS_B/MSE
A×B	$(a-1)(b-1)$	SS_{AB}	MS_{AB}	MS_{AB}/MSE
Error	$ab(r-1)$	SS_E	MSE	
Total	$abr-1$	SS_T		

Each numerator is an extra sum of squares for a well-defined model subspace.

Exact F tests in the balanced two-way model

COCHRAN

Because the model pieces are orthogonal, Cochran’s theorem gives independent quadratic forms.

Null	Numerator df	F statistic
No A effect	$a-1$	MS_A / MSE
No B effect	$b-1$	MS_B / MSE
No interaction	$(a-1)(b-1)$	MS_{AB} / MSE

Under the corresponding null, each numerator divided by σ^2 is central χ^2 with its rank as df, independent of SSE.

Interaction changes how main effects should be interpreted

HIERARCHY

If interaction is negligible

The additive model is scientifically coherent. Main effects summarize stable differences averaged over the other factor.

If interaction is important

A single marginal main effect can hide different—or opposite—simple effects across levels of the other factor.

Hierarchical modeling principle When an interaction is retained, retain the corresponding lower-order main effects even if their individual p-values are large.

Two-way tests are nested-model comparisons

NESTED MODELS

Example: test the interaction.

Reduced model

Additive: $\mu + A + B$

Full model

$\mu + A + B + A \times B$

Difference

$q = (a-1)(b-1)$ new directions

$$F = \frac{(SSE_R - SSE_F) / q}{SSE_F / df_E}$$

This same formula works whether or not the design is balanced—as long as the reduced model is genuinely nested in the full model.

03

Unbalanced designs and the general linear model

When orthogonality disappears, hypotheses—not labels—must drive the analysis.

What changes when the design is unbalanced?

UNBALANCED

- Group or cell sample sizes differ, so factor subspaces are generally not orthogonal.
- There may be no unique additive decomposition of corrected total SS into separate “A” and “B” pieces.
- The amount attributed to a term can depend on which other terms have already entered the model.
- The residual SSE for a specified full model is still unique.
- Nested-model extra sums of squares remain well defined and are often the clearest route to inference.

Loss of orthogonality is the central issue—not “ANOVA versus regression.”

Type I, II, and III sums of squares are different hypotheses

UNBALANCED SS

Type	Core idea	Key caution
I	Sequential: add terms in a specified order	depends on order
II	Test main effects adjusted for other main effects	usually excludes containing interactions
III	Test each term adjusted for all other terms	interpretation depends on estimability/coding conventions

Do not choose a “type” mechanically. State the scientific null hypothesis, then construct the corresponding reduced and full models.

Empty cells create estimability problems

ESTIMABILITY

If some A×B combinations are unobserved, the saturated cell-means model no longer identifies every μ_{ij} .

$$\text{estimable } c^T \beta \iff c \in \mathcal{C}(X^T)$$

Still possible

Many averages and contrasts across observed combinations remain estimable, depending on the design.

Not automatically possible

A nominal “main effect” may involve cell means that cannot be separated from interaction structure or missing combinations.

The estimability question should be checked before interpreting a reported sum of squares or coefficient contrast.

The general projection-matrix template

GENERAL THEORY

Let the reduced and full model spaces satisfy

$$\mathcal{C}(X_R) \subset \mathcal{C}(X_F), \quad q = \text{rank}(X_F) - \text{rank}(X_R)$$

$$SS_{\text{extra}} = Y^T (H_F - H_R) Y$$

$$SSE_F = Y^T (I - H_F) Y$$

$$F = \frac{SS_{\text{extra}} / q}{SSE_F / (N - r_F)} \sim F_{q, N - r_F} \quad \text{under } H_0$$

An ANOVA table is compressed model-comparison bookkeeping

INTERPRETATION

Source

a hypothesis / added model subspace

df

dimension of that added subspace

SS

squared length of the projection into that subspace

MS

SS divided by its dimension

F

signal mean square / independent error mean square

Thinking this way makes the table transferable to regression, ANCOVA, factorial models, and arbitrary general linear hypotheses.

Scope: fixed-effects ANOVA versus random effects

SCOPE

Fixed effects — this lecture

Factor levels are the specific levels of interest. Inference concerns means, contrasts, and finite-dimensional linear hypotheses.

Random effects — different theory

Levels are viewed as sampled from a population. Parameters include variance components, and expected mean squares / mixed-model methods become central.

The same linear-algebra language remains useful, but the covariance model is no longer simply $\sigma^2 I$ after integrating over random effects.

Common mistakes to avoid

PITFALLS

1. Treating ANOVA as unrelated to regression

Both are the same general linear model with different design matrices and hypotheses.

2. Interpreting nonidentifiable effects literally

In effects parameterizations, constraints identify coordinates; estimable functions carry invariant meaning.

3. Memorizing df without rank

Degrees of freedom are dimensions of projection subspaces.

4. Ignoring interaction

Main effects can be poor summaries when interaction is substantial.

5. Using SS “types” without stating the null

In unbalanced designs, the model comparison determines the scientific meaning.

Practice: translate ANOVA questions into linear-model language

PRACTICE

- In a one-way design with $g = 5$ groups and $N = 40$ observations, what are $\text{rank}(H-J)$ and $\text{rank}(I-H)$?
- Write a contrast that compares the average of groups 1 and 2 with the average of groups 3 and 4.
- Why is the effects model $\mu + \alpha_i$ rank deficient, while the cell-means model μ_i is full rank?
- In a balanced 3×4 factorial design with $r = 2$ replicates per cell, give the df for A, B, $A \times B$, and error.
- What reduced and full models would you compare to test interaction?
- Why can Type I sums of squares change when the order of terms changes in an unbalanced design?

Takeaways

SUMMARY

ANOVA model

Factors are encoded through columns of X ; fitted cell means live in $C(X)$.

Estimability

Rank-deficient effects parameterizations require attention to invariant linear functions.

Sums of squares

Each meaningful source is a quadratic form associated with a projection or nested-model increment.

Degrees of freedom

$df = \text{rank} = \text{dimension of the corresponding subspace.}$

Exact inference

Normality + orthogonal/nested projections + Cochran's theorem produce F tests.

Unbalanced designs

State the hypothesis first; then define the reduced and full models that test it.

ANOVA = structured mean spaces + orthogonal geometry + general linear hypothesis testing